

COMPANY PROFILE 2026

AUTONOMOUS SURVEY · UNDERWATER INSPECTION · ENVIRONMENTAL MONITORING · IOT & CUSTOM TECHNOLOGY

WWW.MAKI.NZ · WAIKATO, NEW ZEALAND

INSIDE

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ABOUT US

We go where it's hard to measure

MĀKI is a New Zealand engineering technology company, founded in 2020 and based in the Waikato.

Our team of six designs, builds and operates autonomous systems — uncrewed survey boats (USVs), underwater ROVs, ground rovers, drones and IoT sensor networks — and turns the data they collect into answers for councils, utilities, contractors and researchers.

Most companies are strong in hardware *or* software. We do both in-house: mechanical, electronic and communications engineering alongside our own control software, data platforms and AI. That's why we can build the right tool for a job rather than make the job fit the tool.

Our mission is to make cutting-edge engineering technology accessible to the organisations that manage New Zealand's natural and built environments — capturing data safely, efficiently and in places people shouldn't have to go.

AT A GLANCE

MĀKI in numbers

2020
Founded, Waikato NZ

6
Engineers, developers & operators

26
Waikato lakes mapped for Waikato Regional Council

11
Stormwater ponds surveyed for Western Bay of Plenty District Council

108 ha
Lake Ngaroto surveyed in 3 days

16
Councils, utilities & partners we've worked with

SHE Pre-Qual
Prequalified for safety, health & environment

Built in NZ
Our vessels and software are designed and built in-house

CAPABILITIES

What we do

From concept to deployment, every MĀKI system is built to work reliably in challenging environments — on the water, under it, and beyond.

01

Autonomous systems

Custom USVs, ROVs, rovers and drones, tailored to the job.

02

Survey & mapping

Bathymetry, topography and topo-bathymetry to centimetre-level RTK.

03

Underwater inspection

Reservoirs, ponds, culverts and structures — no draining, no divers.

04

Monitoring & sampling

Water quality profiling and multi-depth samples from autonomous boats.

05

IoT & data

Sensor networks, telemetry, live dashboards and alerts.

06

Software & AI

Mission control, data platforms and AI cameras at the edge.

OUR FLEET

The right vehicle for the site



Hydromap

Autonomous bathymetry USV



Hydrohub

Sampling, monitoring & bathymetry USV



SludgeScan

Dual-frequency & multibeam sediment USV



MicroScan

Compact sonar boat for tight, shallow sites



DeepView ROV

Underwater inspection with live video



Rover

Ground vehicle for confined spaces



AeroMap

RTK drone for 3D mapping



SkyDrop

Waterproof water-sampling drone

SURVEY & MAPPING

Above and below the waterline



Stormwater pond survey: USV sonar, drone mapping and the finished 3D model.

Multibeam and single-beam sonar, RTK positioning and drone photogrammetry, combined into one accurate picture of the site — processed into depth maps, elevation models and 3D outputs tailored to what you need to decide.

- ✓ Depth & contour maps
- ✓ Digital elevation models
- ✓ Integrated 3D models
- ✓ Sediment depth & volume
- ✓ Cross-sections
- ✓ GIS-ready data

Services: **Bathymetry** · **Topography** · **Stormwater ponds**

INSPECTION · MONITORING · IOT

Eyes and sensors where people can't safely go



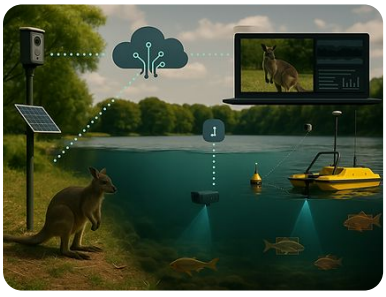
Underwater & confined-space inspection

Our ROV and rover inspect reservoirs, ponds, culverts, outlets and structures with live HD video — no draining, no divers, no confined-space entry.



Water-quality monitoring & sampling

Our sampling vessel profiles the water column with a precision winch: chlorophyll-a, phycocyanin, turbidity, DO, pH and temperature, plus 3 L samples from up to 15 m deep.



IoT & AI cameras

Solar-powered AI cameras detect pest animals such as wallabies at the edge, in remote places with little power or signal. Data flows to live dashboards and alerts.

More: Reservoir inspection · Environmental monitoring · Solutions

PROCESS

How we work

One team from the first call to the final data — no hand-offs between a hardware supplier, a survey crew and a data processor.

01

Scope

We review the site with you, agree the outputs you need, and choose the platform and method that fit — not the one we happen to own.

02

Capture

Our operators deploy the USV, ROV, rover or drone with RTK positioning, working to our SHE Pre-Qual safety system. No divers, no confined-space entry.

03

Process

Sonar, imagery and sensor data are processed in-house into depth maps, sediment volumes, cross-sections and 3D models.

04

Deliver

Clear reports, GIS-ready data and, where it helps, an interactive 3D portal — with the confidence in every figure shown, not hidden.

Safer
Robots go where people shouldn't

In-house
Hardware, software & data under one roof

Fit for site
The method matched to each job

Honest data
Coverage and limits stated plainly

STANDARDS

Safety, quality & biosecurity

Robots take the risk out of the job — and we hold ourselves to the standards councils and utilities expect of any contractor on their sites.

SHE Pre-Qual prequalified

Our safety, health and environment systems are independently prequalified.

No divers, no confined-space entry

ROVs, rovers and USVs go into the water and the tight spaces, so people don't have to.

Biosecurity between sites

IP67-rated, waterproof equipment with a rapid decontamination process between waterways.

Survey-grade positioning

RTK GPS for centimetre-level accuracy on every reading.

Honest about the data

Coverage, methods and limits are stated with every figure we deliver.

Built and supported in-house

We design, build, operate and maintain our own systems — one team accountable end to end.

CASE STUDIES

Real projects, real data

A selection of recent projects for New Zealand councils, utilities and contractors.

> These case studies show some of our work, not all of it — many other projects for clients across New Zealand aren't featured here.

01 **Western Bay of Plenty District Council**
11 stormwater ponds, Ōmokoroa · sonar, RTK & 3D portal

02 **Buller District Council**
Drinking-water reservoir, Westport · ROV & rover

03 **Auckland Council · Glasgow Contractors**
Hobsonville Point jetty · multibeam USV & RTK drone

04 **Waikato Regional Council**
26 lakes mapped · algae-bloom sampling

05 **Ōtorohanga District Council**
Aotea Harbour topo-bathymetry · settling ponds

06 **Lake Ngaroto, Waipā**
108 ha bathymetry & side-scan in 3 days

07 **Dutton Stormwater · Maraetai Power Station**
Stormwater ROV inspection · side-scan sonar

CASE STUDY · WESTERN BAY OF PLENTY DISTRICT COUNCIL

Eleven stormwater ponds, Ōmokoroa

The council needed to know how much water and sediment its Ōmokoroa stormwater ponds and wetlands hold, to plan and prioritise maintenance. Between June and August 2026 MĀKI surveyed eleven of them, matching the method to each site: multibeam sonar, sonar boat surveys, and RTK staff shots where a pond was too shallow or overgrown for a boat.

Each pond was delivered as a survey report plus a private online portal: a 3D model of ground, water and sediment, with a cross-section tool the council's engineers can use themselves. Every pond also shows how much of it lies close to a real measurement — so they know how far to trust each figure.



11
ponds & wetlands

3
survey methods

3D
portal with a cross-section tool

INTERACTIVE DEMO

A stormwater pond in 3D



Ground
banks & surrounds, drone-mapped

Water
surface area & volume held

Sediment
depth, spread & volume

Confidence
coverage shown for every pond

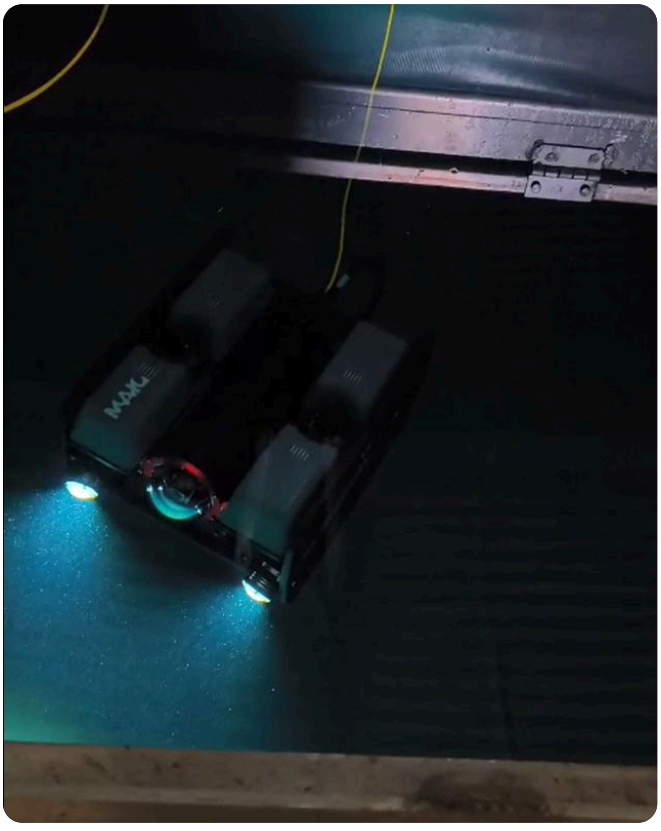
Demo data: the pond bed and depths in this model are illustrative, not survey results. Clients receive full-resolution models and figures in their own private portal. Read the case study →

CASE STUDY · BULLER DISTRICT COUNCIL

Inside a drinking-water reservoir, Westport

Inspecting a drinking-water reservoir normally means taking it out of service: isolating it, draining it, gas-testing the air and sending someone into a confined space.

For Buller District Council, MĀKI inspected the reservoir in Westport with the water left in it. Our ROV surveyed the inside of the reservoir underwater, while our rover carried out the inspection on top of the reservoir — together giving the council a full visual record of the asset.



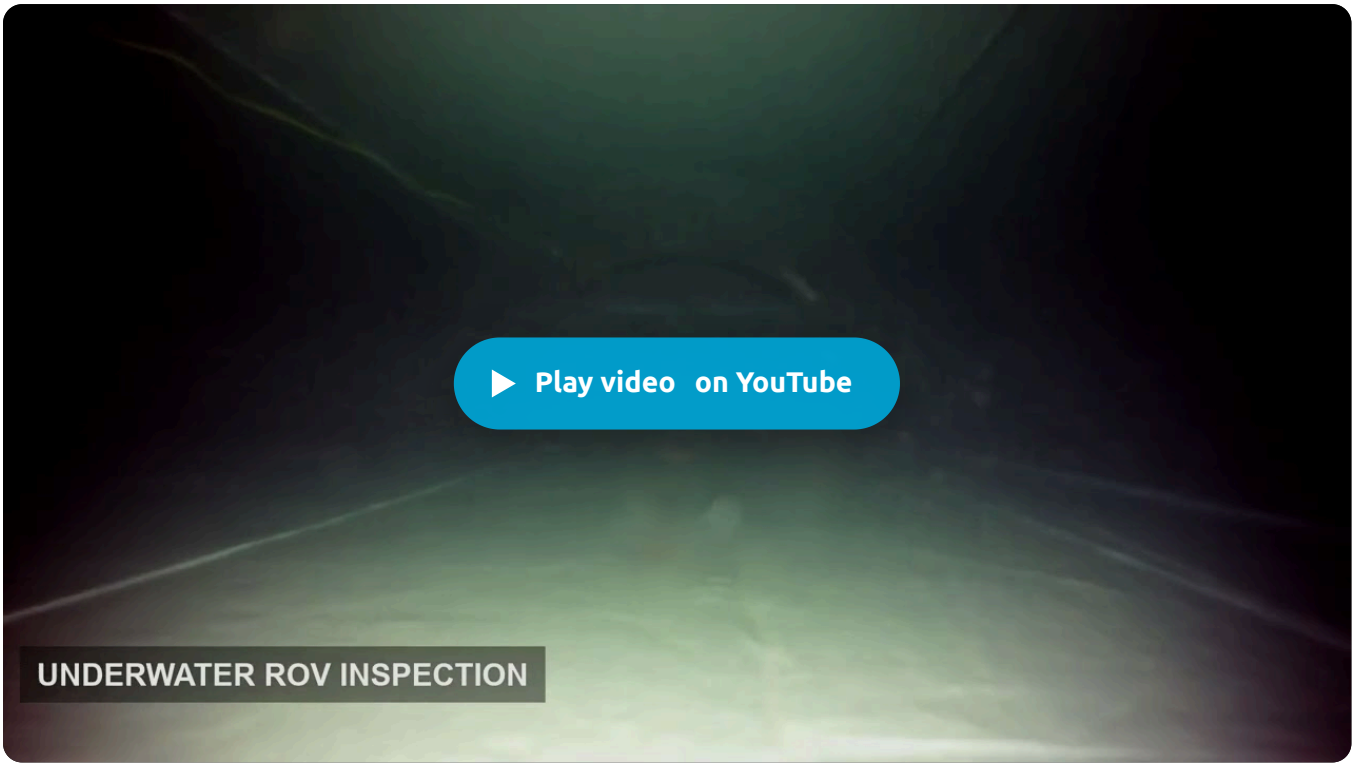
The ROV inside the reservoir



The rover, used on top of the reservoir

BULLER DISTRICT COUNCIL · WESTPORT

Watch the inspection



No draining
Inspected with the reservoir full

No confined-space entry
Nobody had to go inside

Inside and on top
ROV underwater, rover on the structure

Recorded footage
A visual record for condition reporting

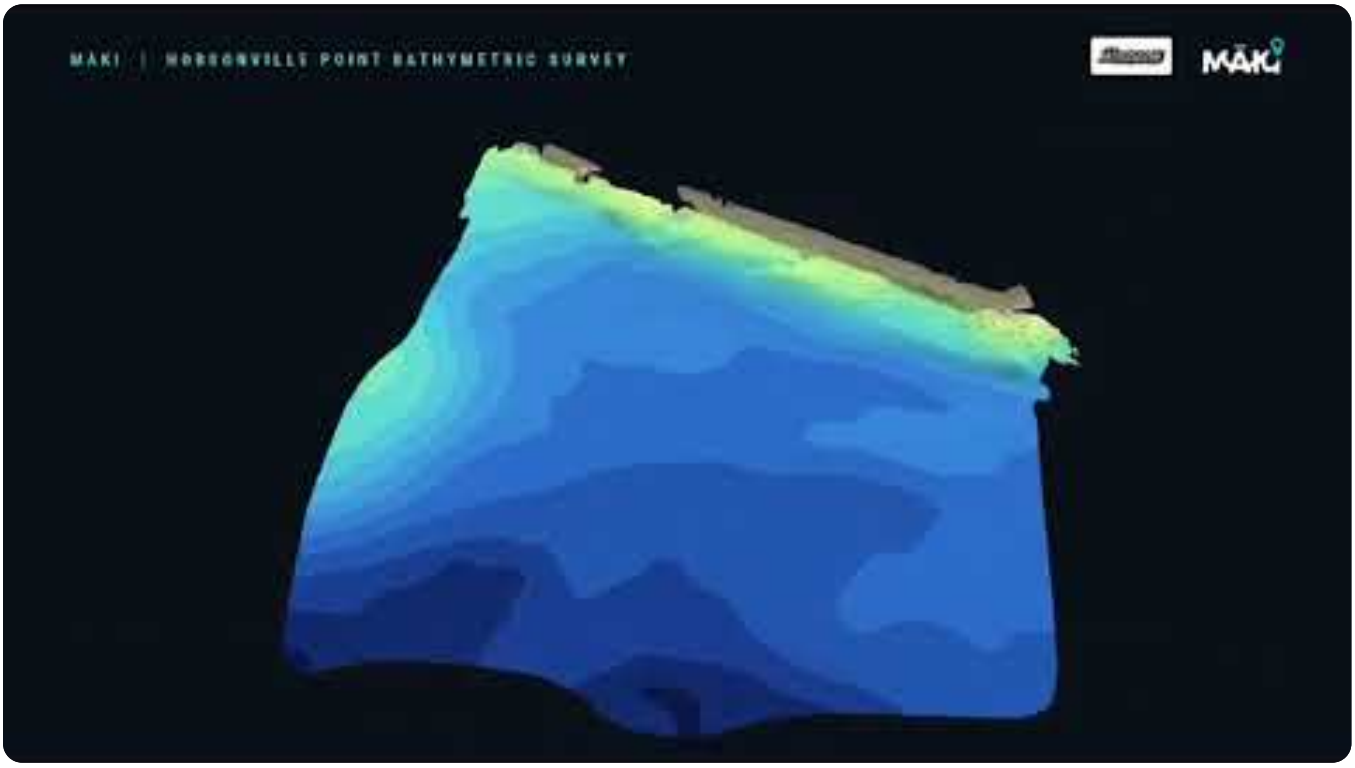
Read more: Full case study · Reservoir inspection service

CASE STUDY · AUCKLAND COUNCIL · GLASGOW CONTRACTORS

Under the jetty at Hobsonville Point

With Glasgow Contractors, on behalf of Auckland Council's Healthy Waters and Flood Resilience department, MĀKI completed a detailed bathymetric and topographic survey at Hobsonville Point on a tight timeframe.

Our pilot drove MĀKI's in-house USV, fitted with multibeam sonar, underneath the jetty — a confined space that crewed survey boats struggle to reach. An RTK drone captured the exposed coastal structures and surrounding land, and the two datasets were combined into one integrated 3D model.

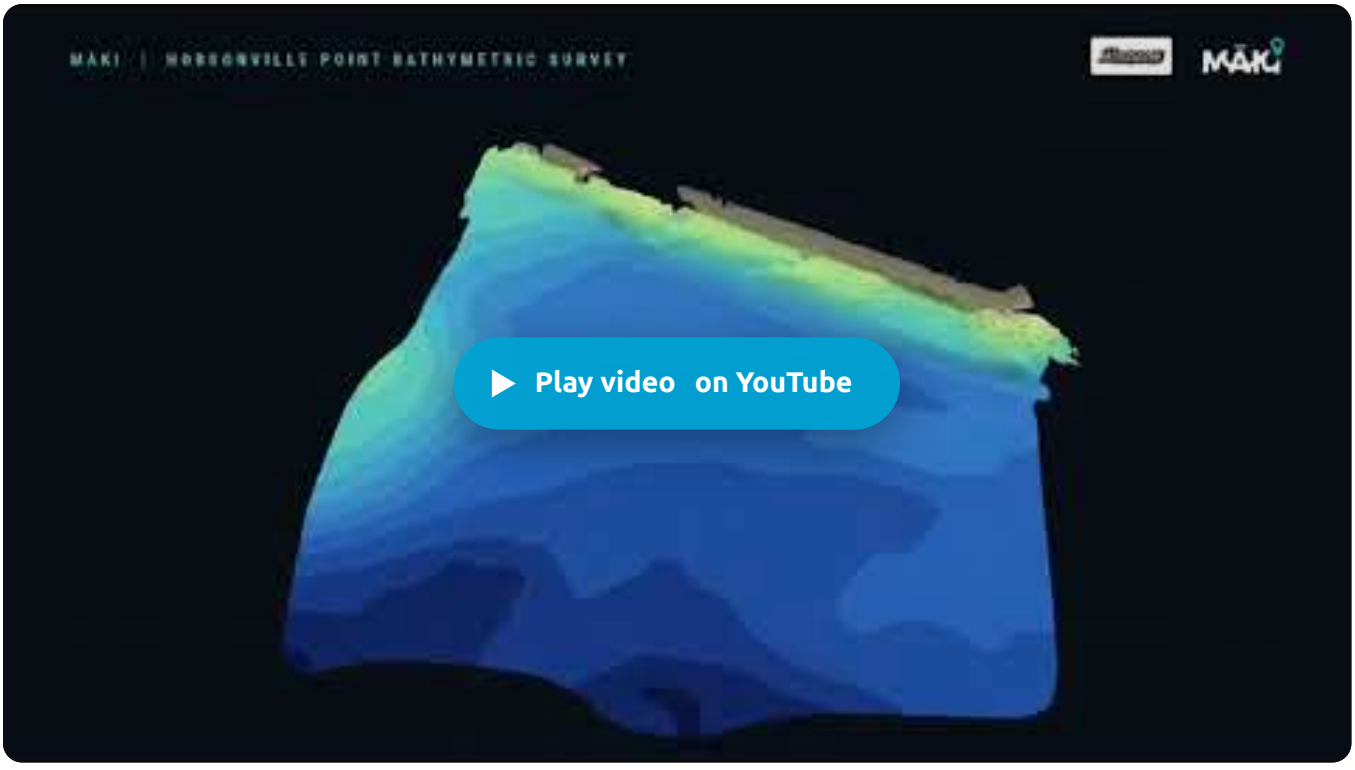


Multibeam seabed model, coloured by depth.

- ✓ Multibeam sonar USV
- ✓ RTK drone photogrammetry
- ✓ Under-jetty surveying
- ✓ Integrated 3D model

AUCKLAND COUNCIL · HOBSONVILLE POINT

Watch the survey



Under the jetty

Data where crewed boats struggle to reach

Above + below

Sonar and drone data in one dataset

3D model

The structure and its surroundings, together

Fast

Mobilised quickly for a tight timeframe

Read more: **Full case study** · **Bathymetry service**

26 lakes, and a safer way to sample blooms



Bathymetry of 26 lakes

High-resolution depth surveys across 26 Waikato lakes with autonomous boats carrying multibeam and single-beam sonar — supporting sediment analysis, habitat mapping and long-term lake management.

Case study →

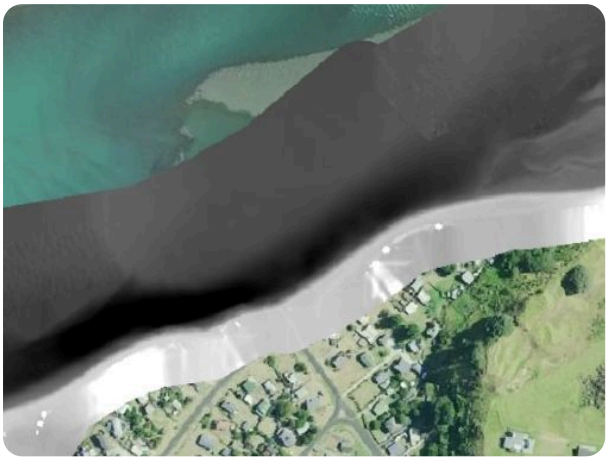


Algae-bloom monitoring

During cyanobacteria blooms our autonomous boat measured six parameters, including phycocyanin, across whole lakes and collected samples at set depths by winch — with no staff on the water.

Case study →

Harbour and settling ponds



Aotea Harbour topo-bathymetry

USV sonar below the water and drone photogrammetry above it, merged into one 3D dataset — including the geometry of the harbour's stone wall for engineering use.

Case study →



Settling-pond sediment volumes

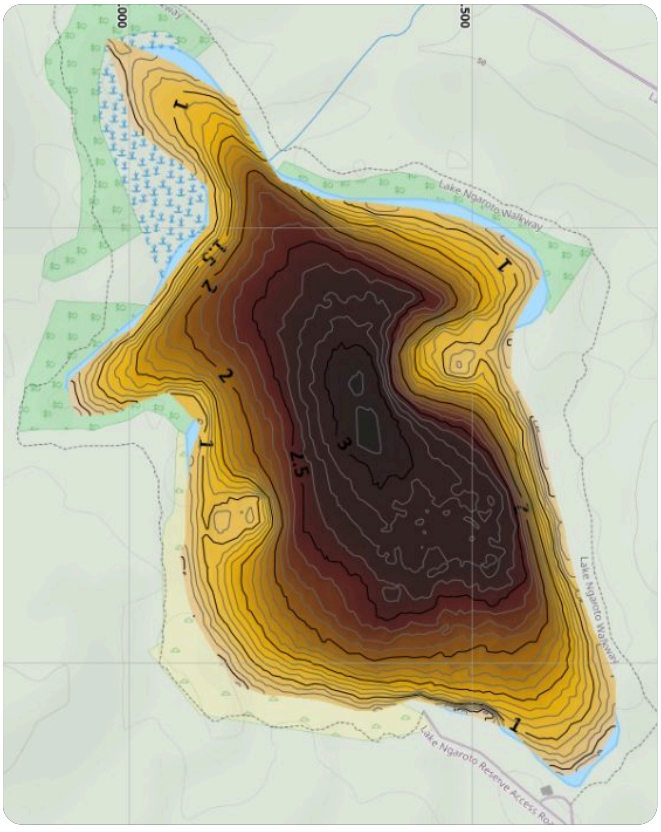
Sediment scanning of settling ponds in Ōtorohanga town centre, giving the council sediment volumes and 3D models to plan maintenance before capacity runs out.

About settling-pond mapping →

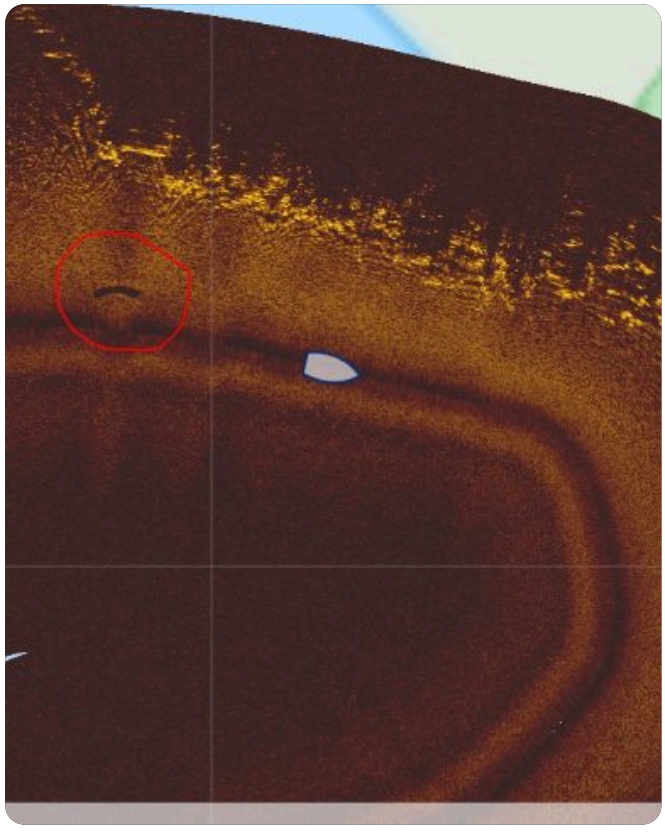
CASE STUDY · LAKE NGAROTO, WAIPĀ

108 hectares in three days

A full bathymetric and side-scan sonar survey of Lake Ngaroto from MĀKI's autonomous inflatable e-kayak — producing a complete depth map, a 3D lake model and a side-scan mosaic of the lake floor.



Depth map (contours in metres)



Side-scan sonar of the lake floor

108
hectares surveyed

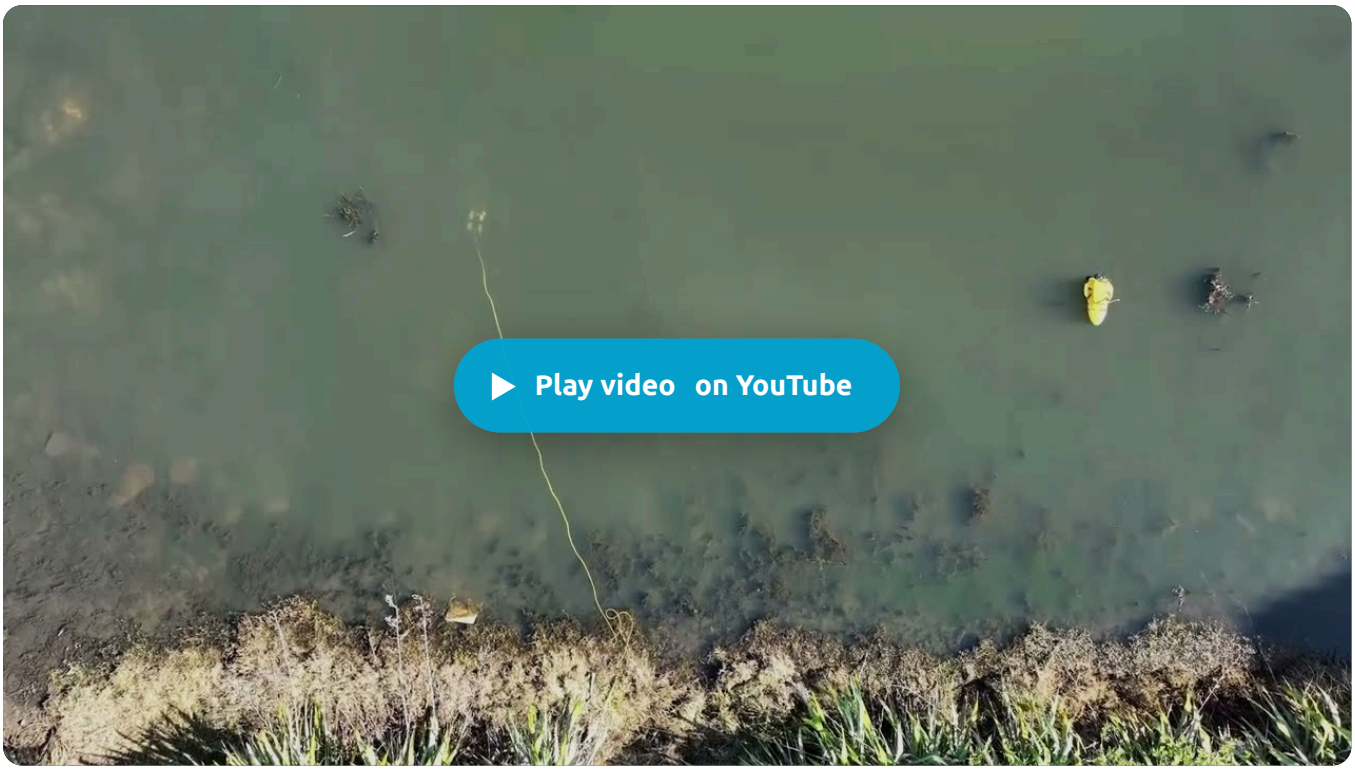
3
days on site

55 m
unknown submerged structure found

Full case study →

MORE PROJECTS

Stormwater ROV and a power-station canal



Underwater stormwater inspection · Dutton Stormwater

Working alongside Dutton Stormwater, MĀKI used an ROV to inspect submerged stormwater infrastructure without anyone entering the water — recording footage for further assessment. The same approach suits ponds, culverts and outlets.

Side-scan sonar · Maraetai Power Station

A side-scan sonar survey of the station's inlet canal captured detailed imagery of sediment layers and underwater structures to support inspection and maintenance planning.



RESEARCH & COLLABORATION

Field-proven, scientifically validated

We work with regional councils, universities and industry partners to develop and validate new technology for environmental monitoring and automation.

Through partnerships with organisations such as Waikato Regional Council and Auckland University of Technology, our engineering and software teams contribute to studies in autonomous sampling, IoT integration and AI-based sensing.

We welcome new research and development partnerships that combine scientific insight with engineering innovation.

News: **MĀKI at Fieldays 2025** · **Freshwater Conference 2025**

DEVELOPMENT ROADMAP

What we're building next



3-in-1 integrated platform

A USV that acts as an "aircraft carrier" for a drone and an ROV. Solar and hybrid power keep all three charged for long missions; the drone returns to its dock to recharge, and the ROV deploys from the boat's central bay — operated wirelessly, with no one needed on the vessel.



Submerged pipe inspection & 3D modelling

A robotic crawler for wet or submerged pipes, with LiDAR, sonar and HD cameras. SLAM positioning works without GPS to build 3D point clouds and digital twins for defect finding and asset management.

CLIENTS & PARTNERS

Who we work with

Waikato Regional Council — Te Kaunihera ā Rohe o Waikato

Western Bay of Plenty District Council

Buller District Council — Te Kaunihera O Kawatiri

Auckland Council — Te Kaunihera o Tāmaki Makaurau

Watercare

Te Kaunihera ā-rohe o Ōtorohanga District Council

Mercury

Glasgow Contractors

Dutton Stormwater

Vernon Developments — Cartage, Earthmoving, Aggregate Recycling

Conhur — the new organic

Te Wānanga o Aotearoa

Submerged — Environmental & Scientific Dive Services

deepdive division

Torpedo7

PAKnSAVE

CONTACT US

Let's talk about your next project

We welcome enquiries from councils, utilities, contractors and research institutions — for a one-off survey, an ongoing monitoring programme, or a custom system.

Get a quote

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